

Mohammad Labak

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[GitHub](#) | [My Website](#) | [LinkedIn](#)

SOFTWARE ENGINEERING CO-OP STUDENT

Computer Science (Co-op) student specializing in **software systems, automation, and manufacturing-focused applications**. Strong background in **system integration, CI/CD pipelines, data-driven operations, and 3D technical design**.

SKILLS

Languages: Python, C++, Java, JavaScript

Frameworks: Flask, Node.js, React, Next.js

Databases: MongoDB, MySQL

DevOps & Tools: Git, Docker, Linux, REST APIs, Stripe, Vercel, Railway

Engineering & Design: AutoCAD, Onshape, 3D CAD Modeling

Core Areas: Automation, System Integration, CI/CD, Backend & Full-Stack Development

PROJECTS

Sentinel FieldView - Real-Time Navigation & Coordination System

C# · .NET · WinUI 3 · UDP

- Built a system where a central operator can guide multiple remote users in real time using a shared floorplan with live visual overlays (paths, zones, markers).
- Developed an interactive desktop interface for creating and editing these overlays, with changes appearing instantly on connected devices.
- Engineered a custom UDP-based networking layer to keep all clients synchronized with low latency, even under packet loss or unstable connections.
- Designed the system around an authoritative relay model, ensuring consistent shared state across all users and preventing conflicts between updates.

Manufacturing Operations & Ordering System - Allpac Container Ltd.

React · Node.js · MongoDB · Stripe

[View live site](#)

- Built a full-stack ordering and pricing system for a packaging manufacturer managing 12 production SKUs with **secure B2B payment processing**.
- Designed backend workflows to **link orders directly to production constraints**, enabling accurate pricing, inventory tracking, and repeat customer ordering.

Real-Time Speech Processing & Automation Tool - FixUrSpeech AI

Python · Flask · JavaScript · Machine Learning APIs

- Built a real-time speech processing pipeline for recording, transcription, correction, and audio playback.
- Designed a modular backend architecture for **low-latency machine-learning inference and real-time debugging**.

- Integrated asynchronous processing and logging to **monitor pipeline stages in real time** and isolate performance bottlenecks during live audio sessions.

Product Data Management Platform - LaPurity Water Tech

Next.js · Sanity CMS · Node.js

[View live site](#)

- Built a structured product catalog system for industrial water treatment equipment using a headless CMS.
- Implemented automated data sync and optimized queries to achieve 95+ Lighthouse performance.
- Structured content schemas and API queries to **support scalable product expansion** without requiring frontend code changes.

WORK EXPERIENCE

Software Engineer Intern

Jan 2026 – Present

Allpac Containers Inc., Windsor, Ontario

- Building a full-stack ERP system (React, Node.js, Express, PostgreSQL, Prisma) to handle inventory, production, purchasing, and warehouse workflows.
- Designing backend APIs and core system logic for inventory transactions, work orders, and product data, with role-based access and proper state validation.
- Working closely with the operations team to turn real warehouse processes into structured, reliable software, improving traceability and reducing manual work.

Teaching Assistant

Sep 2024 – Dec 2025

University of Windsor, Windsor, Ontario

- Guided students through Python & C++ labs, debugged code live, and graded assignments with structured feedback.

Design assistant (Co-op)

May 2022 - Aug 2022

MMA Architect, Windsor, Ontario

- Created **precision 3D technical drawings and layouts** in AutoCAD to support architectural design workflows.
- Developed strong spatial reasoning and engineering visualization skills.

EDUCATION

Bachelor of Computer Science (Honours, Co-op)

University of Windsor (3rd Yr)

2023 - Present

- Dean's List

Relevant Coursework: Software Development, Software Engineering, Databases

CERTIFICATIONS

Career Essentials in Generative AI - Microsoft

Dec 2024

Career Essentials in Software Development - Microsoft

Nov 2024